DCLAB · OFFICIAL DOCUMENT

The Plan to Reach the Vision

> **Six phases, from the first real data teams to each company's own decision model.**

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1. Summary

DCLab has built a strong base over thirteen phases: an engine for most kinds of problem, leakage checks, traceable results, a record of every decision, and agents measured on sealed exams. All of it was measured on written people and invented or public tables. **No real data team uses DCLab every week yet, and no company has sent real outcomes back.** The vision is about real companies, their decisions and their outcomes, so the plan starts there.

Phase	Goal	Done when
14 Real data teams	A Python SDK and three to five real teams using DCLab on their own problems.	Three teams use it weekly for four weeks; caught flaws and time saved are measured.
15 One model, one decision	A decision card for managers, built from pieces that already exist.	On three real problems, the promised gain is later checked against the outcome.
16 The live loop	Models in use: decisions logged, outcomes collected, monitoring and retraining.	One company completes two cycles; the second model beats the first on newer data.
17 Many models, one decision	The company decision model: several models, shared budgets and rules, one action.	It beats each model's own rule in measured gain: on an invented table, then in one company.
18 Grows with each company	Company memory and the company's own small models.	The company's own model beats the general one on new decisions, without forgetting.
19 The service	Cloud for many companies, security, pricing and sales.	Paying customers.

2. Purpose and scope

This document turns the vision into an order of work. It says what is missing, which phase closes each gap, how each phase is judged, and what the owner must decide. It is a plan, not a design: each phase will get its own detailed plan

with numbered packages (from 38.x onward), written when the phase starts, in the same way as phases 1 to 13.

Phase numbers continue after phase 13 (Claude first, the lead agent). Every phase follows DCLab's usual method: a test that fails first, the change, the checks, an adversarial review, a development record, and a gate fixed before the measurement.

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3. The starting point

What already exists and will be reused. Paths are in this repository.

Capability	Where	State
Problem engine: 15 kinds supported, 2 partial, images not yet	`dclab_rnd/kinds/`, `dclab_rnd/studio/`	Built
Leakage review, train-only exploration, the final test read once	`studio/leakage_review.py`, workflow graph WF-01 to WF-10	Built
Code export that rebuilds the final number in a clean environment	`studio/code.py`, `make code-check`	Built
Lineage: every number traced to its run	`dclab_core/lineage.py`, `studio/lineage.py`	Built
Operating point from costs and value per case; cost matrix	`dclab_core/report.py`, `solution_brief.py`	Built, one model
List sized to the team's capacity; uplift	`dclab_core/capacity.py`, `uplift.py`, `anomaly.py`	Built, one model
Prediction, decision and outcome log	`dclab_core/serving.py`, `outcomes.py`	Record shapes built; never fed by a company
Monitoring and retraining (champion and challenger)	`dclab_core/monitoring.py`, `retrain.py`	Built; never run in use
Decision ledger (45 points), consent, redaction, storage	`dclab_core/ledger.py`, `dclab_rnd/consent.py`, `store/`	Built
Small decision models in shadow, promoted only when measured better	`dclab_rnd/ledger/routing.py`, `model_version.py`	Built
The lead agent (understands first)	Phase 13	Partly proven: understanding 0.93 against 0.82 for Claude alone; failed on route (0.79 against 1.00) and traps
Open-source studio `dclabai`; developer app DCLab Code; tester kit	public repository; private repository; `dclab_rnd/testers/`	Built

Capability	Where	State
Cloud deployment code (AWS, Google Cloud)	`deploy/`	Written; never applied

4. Gap analysis

Each gap is something the vision needs that does not exist or has not been proven. The last column says which phase closes it.

#	Gap	Why it matters	Closed in
G1	No real team uses DCLab weekly.	Every later phase needs real data, real decisions and real outcomes. Without users, every measurement stays on invented cases.	14
G2	No Python SDK. A data scientist cannot write `import dclab` in a notebook. `dclabai` is a whole studio, not a small library.	The notebook is where data teams work. It is the easiest door into a company.	14
G3	The lead agent is not proven: it failed its sealed gate on choosing the route and on traps.	The first thing a new user meets is the agent. A wrong route loses trust on day one.	14
G4	No business view. Pages are made for data scientists.	The decision-maker needs actions, gains and reasons, not metrics.	15
G5	Promised gains are never checked against what happened.	Without this check, a decision card is an opinion, not evidence.	15
G6	No path to use. The export rebuilds the score but does not run as a scheduled job inside a company.	A model that is not used produces no decisions and no outcomes.	16
G7	The outcome loop never ran. Outcomes also arrive late (churn is known weeks later).	"Grows with the company" means learning from outcomes. No outcomes, no growth.	16
G8	Learning bias: acting only on the cases the model picks teaches it nothing about the others.	The model would learn only from its own choices and could never correct its blind spots.	16
G9	Many models are never combined into one decision.	This is the heart of the business vision.	17
G10	No way to share budgets, capacity and rules across models, or to settle models that disagree.	Real decisions are limited by shared resources.	17
G11	No company has its own trained model.	The company-owned model is the lasting asset the vision promises.	18
G12	No multi-company cloud, security review, pricing or sales.	Needed to become a business, not a research project.	19

#	Gap	Why it matters	Closed in
G13	No decision on who buys first: data teams or business buyers.	Two buyers at once can split a small team.	Owner decision D2

5. The six phases

Sizes are relative (S, M, L), not dates. Dates are set when each phase is planned in detail.

Phase 14 · Real data teams Size L

Put DCLab in the hands of three to five real data teams, through a Python SDK in their own notebooks, and prove the lead agent.

Why now

Every other phase depends on real users (G1). This is also the cheapest phase to learn from.

Builds on

The engine, `dclab_core` (portable, installs as a wheel), the notebook companion, the tester kit, the open core `dclabai`, the lead agent of phase 13.

Work

- **The SDK.** A small library over `dclab_core`: check a table or a notebook for leakage and split errors; build a model with the engine; export the code. Three or four calls, no server needed.
- **First teams.** Three to five teams on their own problems, running locally with the tester kit, so their data never leaves their machine. A short weekly check-in with each team.
- **Prove the lead agent.** Fix the route and trap failures on practice exams only, then one fresh sealed exam. Until it passes, Home keeps its rules.
- **Capture.** With each team's consent, record sessions, decisions and corrections, as every phase does.

Out of scope

The decision layer, the cloud, pricing.

Done when

Three teams use DCLab every week for four weeks. In each team, at least one real flaw (leakage, a wrong split, a false score) is caught before export and confirmed by the team. Time from raw data to the first trusted model is measured. The lead agent passes a fresh sealed exam, or the result is recorded and Home stays on rules.

Risks

Finding teams willing to try it; privacy concerns (answered by running locally); the SDK growing too large (keep it to a few calls).

Owner decides

D1 (which teams), D3 (SDK open or private).

Phase 15 · One model, one decision Size M

Turn one model's predictions into a decision card a manager can act on, and check its promise later.

Why now

The pieces exist, so it is cheap, and it is the first thing a decision-maker can see (G4, G5).

Builds on

The operating point from costs and value per case, the cost matrix, capacity-sized lists, uplift, the solution brief's success bar, outcomes linked to decisions.

Work

- **The decision card.** Who to act on and how many; the expected gain with its range; the reasons, every number traced; what changes if a cost or a budget changes.
- **The manager's page.** One simple page with the card, separate from the data scientist's pages.
- **The promise as a record.** Each card stores its expected gain as a decision record, so the real result can be compared with it later.

Out of scope Several models together; automatic scheduling.

Done when On three real problems from phase 14's teams, a decision card was used, and its promised gain was later compared with the observed result, whatever that result was.

Risks Teams cannot state their costs (offer ranges and show how the decision changes across them); outcomes take long (choose problems with short feedback first).

Phase 16 · The live loop Size L

Run a model in use inside one company: score, decide, log, collect outcomes, watch, and retrain.

Why now Phases 17 and 18 cannot be judged without real outcomes (G6, G7, G8).

Builds on The code export, the prediction-decision-outcome log, monitoring, the retrain policy with champion and challenger, lineage.

Work

- **Batch scoring first.** The exported pipeline runs as a scheduled job (daily or weekly) inside the company. A live API comes later, only if needed.
- **Outcomes in.** A simple way to bring outcomes back: a file or a database table, matched to decisions by key.
- **Watch and retrain.** Monitoring alerts; retraining only when enough new labelled rows exist; the challenger replaces the champion only if it wins on the newest window.
- **A small exploration share.** About 5% of cases, agreed with the company, get a different action, so the model keeps learning about the cases it would not pick.
- **Promise against reality.** A regular report: what each decision card promised, and what happened.

Out of scope Real-time serving; many companies at once.

Done when One company completes two full cycles (score, decide, outcome, retrain), and the second model beats the first on newer labelled data, with its range. The exploration share is in place and agreed.

Risks Access to the company's systems; outcome delay; the exploration share has a cost (keep it small, agreed and visible).

Owner decides D4 (exploration policy), D5 (the first company).

Phase 17 · Many models, one decision Size L

Build the company decision model: several models about the same customer or case, under shared budgets, capacity and rules, turned into one best action.

Why now	It is the heart of the vision (G9, G10), and it can be judged only once phase 16 brings real outcomes.
Builds on	Entity tables (one row per customer at a cutoff), uplift, capacity, the composition of several leaves into one programme, the decision card.
Work	• **One view per case.** Every model's prediction about the same customer or order in one place. • **Constraints.** Budgets, capacity, business rules and minimums, written once and shared by all models. • **Choosing.** An optimiser that picks the actions with the highest expected value under the constraints. • **Disagreements.** Clear rules when models disagree, and the reason shown. • **Invented table first.** A replica with planted truth (for example churn, value and uplift for a telecom company), so the right answer is known; then one real company.
Done when	On the replica, it beats each model's own rule by a margin fixed before the run. In one real company, its measured gain beats the previous way of deciding, with its range.
Risks	Uplift needs history where actions varied (phase 16's exploration share provides it); too many constraints make results hard to explain (start with budget and capacity only).

Phase 18 · Grows with each company Size M

Each company gets its own memory and its own small models, trained on its own decisions and outcomes.

Why now	This turns DCLab from a tool into the company's lasting asset (G11). It needs months of decisions and outcomes from phases 16 and 17.
Builds on	The agents' memory (phase 11), the small decision models in shadow and their routing (phase 7), versions with roll-back and the no-forgetting exam, consent.
Work	• **Company memory.** Its costs, rules, corrections and past decisions, read by every agent. • **Company models.** Small models trained on the company's own decisions and outcomes, run in shadow first. • **Promotion.** A company model takes over a choice only after it measured better than the general one on the company's new decisions. • **No forgetting.** A frozen exam per company, run before each new version is accepted.
Done when	After several cycles, the company's own model beats the general one on its new decisions, and passes its no-forgetting exam.
Risks	Too little data per company (pool across companies only with consent, never their values); a model that fits the past too closely (judge only on newer data).

Phase 19 · The service Size L

Offer DCLab to many companies, safely, at a price.

Why now	Only after phases 14 to 16 show value in real use (G12).
Builds on	The cloud code in `deploy/`, accounts and roles, usage limits, the audit trail, consent and retention.
Work	A multi-company cloud; a security review; where data lives; pricing (for example per data team for layer 1, and by decision value for layers 2 and 3); onboarding and support; the line between the open core and the paid service.
Done when	Paying customers, in a number the owner sets before the phase starts.
Risks	Selling to business buyers takes long; security reviews of large companies take long.

6. Order and dependencies

14 Real teams → **15 One decision** → **16 Live loop** → **17 Many models** → **18 Company models** → **19 Service**

- **17 needs 16.** A combined decision cannot be judged without real outcomes, and uplift needs the variety that the exploration share brings.
- **16 needs 14.** A live loop needs a real company, and the first company comes from the first teams.
- **15 can overlap 14.** Its pieces exist; it can start as soon as the first team has a model.
- **18 needs time.** It starts once a company has several cycles of decisions and outcomes.
- **19 can be prepared early** (security, pricing ideas), but selling waits for measured value.

Why this order, in one picture. A restaurant: first the kitchen must cook well (14), then the menu must suit the guests (15), then guests must come back (16). Only then does a bigger menu (17), a chef who knows each regular guest (18), or a second restaurant (19) make sense.

7. What continues and what pauses

Continues in every phase

- Capture: every agent and process records its full data, with the model's thinking summary, for DCLab's own models.
- The rules of honest measurement: gates fixed before runs, sealed exams read once, promotion only when measured better.
- The lead agent's fixes, until it passes a fresh sealed exam (part of phase 14).

Pauses until phase 14 has real users (proposed)

- New internal exams beyond the lead agent's.
- More small decision models ("students") for new decision points.
- New storing and archive work beyond keeping what exists running.
- Phase 9's learning loop, as a separate phase: its useful parts move into phases 16 and 18.

These are good tools, but they do not move DCLab toward the vision now. Real users do.

8. Risks

Risk	Effect	Response
No team agrees to try DCLab	Phase 14 stalls, and so does everything after it.	Start with people the owner knows; run locally so data never leaves; offer the leakage check as a free first step.
Building infrastructure instead of meeting users	More phases, still no proof.	Each phase ends only on a real-use test; the pause list in section 7.
Two buyers at once	A small team split between two kinds of product.	Data teams first (recommended in D2).
Slow outcomes	Phases 16 to 18 take longer than planned.	Choose first problems with short feedback (days or weeks, not months).
Exploration share refused or too costly	The model learns only from its own picks.	Keep it small, agreed and visible; show what it teaches.
Results tuned on invented cases	Good scores that fail with real companies.	Invented replicas only for a first check; the real test is always a real company.
Company data concerns	Companies refuse to share.	Local runs, consent off by default, table values never in records, the audit trail.
Limited model spend	Fewer live tests than needed.	Small capped runs; local models and rules where they measured good enough.

9. Scoreboard

The numbers to report at the end of each phase.

Phase	Main number	Also report
14	Teams using DCLab weekly	Flaws caught and confirmed; time to first trusted model; the lead agent's sealed result
15	Decision cards used	Promised gain against observed result, per card
16	Full cycles completed	Second model against first on newer data; monitoring alerts; exploration share kept
17	Measured gain against each model's own rule	Result on the replica; result in the company, with its range
18	Companies whose own model beats the general one	No-forgetting exam results
19	Paying customers	How long they stay; the value of their decisions

10. Decisions needed from the owner

#	Decision	Recommendation	Needed for
D1	Which three to five real teams or companies come first?	People the owner knows, with problems that have short feedback.	Phase 14
D2	Who comes first: data teams or business buyers?	Data teams. They are the door into each company, and the decision layer needs their models.	Phase 14
D3	Is the SDK part of the open-source `dclabai`, or private?	Open: it makes the first step free and easy; the decision and learning layers stay private.	Phase 14
D4	The exploration policy: how large a share, and who approves it?	About 5%, agreed with each company and shown in every report.	Phase 16
D5	Is HyperAck a possible first company for the live loop?	Ask, since its data is already in the project (under its terms).	Phase 16
D6	Do you agree with the pause list in section 7?	Yes.	Now

11. Related documents

- DCLab Vision: the goal this plan works toward.
- `docs/guides/ROADMAP.md`: phases 1 to 13, built.
- `docs/guides/THREE_MODELS.md`: DCLab's own models and their data.
- `docs/guides/PHASE9_LEARNING_LOOP_PLAN.md`: the learning loop plan, folded into phases 16 and 18 (proposed).
- `docs/guides/TESTERS.md`: the tester kit used in phase 14.